Bahir Dar Institute of Technology

Faculty of Computing

Wireless Communication and Mobile Computing worksheet

Target Groups: (Cyber & CS) 3rd

Part I: Choose the best possible Answer

1. Which one of the following is true about radio wave?
   A. It has high frequency
   B. It uses unidirectional antenna
   C. It is used in wifi and short range communication
   D. All

2. Which one of the following is not true about 2G?
   A. It uses OFDMA as multiple access technique
   B. It is used for both voice and data communication
   C. It uses digital signal
   D. None

3. Why we prefer counter clockwise direction to determine co-channel cells?
   A. Because it is convenient
   B. Because there is interference from sunlight
   C. Because there is interference from other non-cellular systems
   D. Because there is interference from earth's magnetic field

4. Which one of the following is different from the others?
   A. GSM
   B. PDC
   C. NADC
   D. AMPS

5. Which feature characterizes early telephone system?
   A. Multiple simultaneous communication was possible
   B. Marconi's telegraph is a sophisticated digital telephone
   C. The security was hindered
   D. All

6. Why we prefer hexagon for denoting the shape of the cell?
   A. Because it is geometric
   B. Because there no overlapping and uncovered region
   C. Because all sides are equal and highly discriminant
   D. All

7. Which one of the following is an example of half duplex communication?
   A. Walkie talkie
   B. Printing
   C. Phone call
   D. None

8. Which IEEE standard does Bluetooth uses?
   A. 802.3
   B. 802.11
   C. 802.15
   D. 802.15.4

9. What is the role of MSC?
   A. Connects mobile stations with the base transceiver station?
   B. Routes our communication along with the best alternative path
   C. Connects base station with the central office
   D. Setup the call

10. What will happen if the size of the cluster is very small?
   A. More number of cluster is required to provide full coverage of the zone
   B. Frequency reuse factor will decrease
   C. The system capacity will decrease
   D. None

Part II: Provide the answer Accordingly

1. Differentiate static and dynamic channel assignment strategies with their advantages and limitations.
2. Briefly explain what handoff is and why handoff is necessary.
3. What is antenna and its role? Briefly explain.
4. What we call if I am able to use my ethiotelecom's SIM card in USA as it is? Define it.
5. Differentiate 3G and 4G at least with two criteria.
6. What makes 2.5G preferable over 2G? list at least two good sides of 2.5G.